PROJECT REPORT

California Traffic Collisions Analysis

ETL Pipeline, Data Preparation and Exploratory Analysis

SWITRS collision records | Apache Spark | Exploratory data analysis

Prepared from the analysis workflow and saved outputs of the supplied Jupyter notebook.

Executive Summary

This project applies an Extract–Transform–Load (ETL) workflow and exploratory data analysis to a supplied California traffic-collision extract associated with the Statewide Integrated Traffic Records System (SWITRS). Apache Spark is used to process collision, party and victim tables, standardize analytical fields, profile data quality, and aggregate patterns by severity, weather, road surface, lighting, location and time.

The cleaned collision-level analytical dataset contains 935,764 records. Property-damage-only cases are the largest severity group (570,460; 61.0%), followed by cases labelled pain (226,486; 24.2%), other injury (116,680; 12.5%), and severe injury (22,138; 2.4%). Los Angeles County has the largest number of records (284,088). Friday is the busiest weekday (154,539), October is the highest-count month (83,272), and 17:00 is the most frequent hour (73,252).

These are descriptive record counts, not exposure-adjusted measures of collision risk. The supplied collision-level fatality field is zero throughout non-null records, so a reliable fatal-collision percentage cannot be derived from that field alone.

Project at a Glance

Item	Description
Source	Supplied SWITRS-related CSV extract
Input tables	Collisions, parties, victims and case IDs
Cleaned collision-level records	935,764
Time coverage	2001–2021; final year is partial
Processing	PySpark ETL, quality profiling, aggregation and visualization
Dimensions	Severity, weather, victim age, lighting, road surface, county, weekday, month and hour

1. Introduction and Problem Statement

Traffic collisions affect public safety, healthcare demand, mobility and infrastructure planning. Historical records can help identify where and when incidents are concentrated and how recorded severity varies across environmental and temporal categories. This project develops a repeatable data-processing workflow and summarizes key patterns from the supplied California collision extract.

2. Objectives

- Build a Spark-based ETL workflow for the supplied structured CSV tables.
- Inspect schemas, missing values, duplicate records and numeric outliers.
- Prepare standardized fields for severity, weather, road surface, lighting, dates and times.
- Explore patterns by severity, victim age, lighting, weekday, county, geography and time.

- Answer ETL queries on county, month, weather, hour, road surface and lighting.
- Translate descriptive results into practical road-safety considerations.

3. Data Sources and Scope

The notebook loads four CSV tables: sample_collisions.csv, sample_parties.csv, sample_victims.csv and sample_case_ids.csv. Recorded input sizes are 935,791 collision rows, 1,866,917 party rows and 963,933 victim rows. The cleaned collision-level analytical frame contains 935,764 records. Date values span 2001 through 2021; 2021 is partial in the supplied extract.

4. ETL Methodology

Extract. The notebook mounts Google Drive and reads the four CSV files from the project data folder. Source identifiers are read as strings to preserve their representation during ingestion.

Transform. The workflow inspects schemas and sample records, profiles null and blank values, converts selected numeric fields, parses collision dates and times, standardizes categorical labels, and creates year, month, weekday and hour fields. It checks exact duplicate rows and repeated collision keys and uses interquartile-range screening to flag numeric values for review. Sparse location fields are retained rather than removed automatically.

Load and analyze. The cleaned analytical collision frame supports Spark aggregations and visualizations. The notebook also saves cleaned analytical data in Parquet format for optional reuse. Geographic plotting uses valid coordinate records sampled for visualization, while frequency aggregations use the collision-level analytical frame.

5. Data Quality and Preparation

The notebook reports no exact duplicate collision rows and 27 rows beyond the first occurrence of a case identifier. Missingness is concentrated in location-related fields: location type, Caltrans county and side-of-highway are each missing in about 72.9% of records; latitude and longitude are each missing in about 71.5%. Numeric screening flags values for review, while extreme values are not removed automatically. Victim-age visualization uses values between 0 and 110 years.

6. Exploratory Data Analysis

6.1 Collision Severity

Property-damage-only records dominate (570,460; approximately 61.0%). Records labelled pain account for 226,486 (24.2%), other injury for 116,680 (12.5%), and severe injury for 22,138 (2.4%).

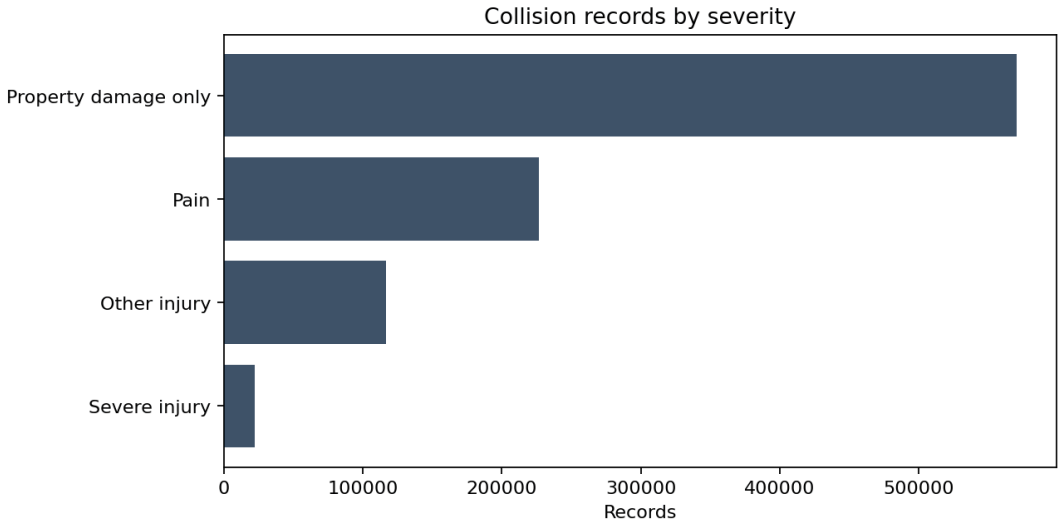


Figure 1. Collision records by severity.

6.2 Weather Conditions

Clear weather is the most frequently recorded condition (769,905), followed by cloudy (122,150) and raining (32,341). Another 4,626 records have no standardized weather value. Raw counts do not imply that clear weather is more hazardous, because exposure under each condition differs.

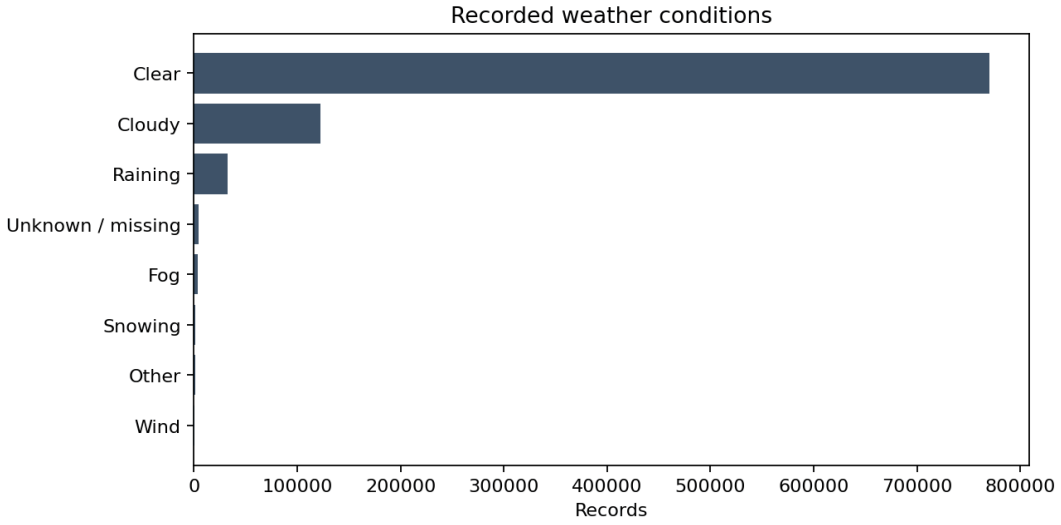


Figure 2. Recorded weather conditions.

6.3 Victim Age and Victim Counts

The notebook examines victim ages using numeric values between 0 and 110 for visualization and compares recorded victim counts by severity. Mean recorded victim counts are approximately 1.37 for pain, 1.51 for other injury and 1.62 for severe injury. These are descriptive summaries based on available source fields.

6.4 Lighting Conditions

Daylight has the largest count (625,553), followed by dark with street lights (195,864), dark with no street lights (74,632), and dusk or dawn (32,646). Missing lighting labels total 5,301; 1,768 records are labelled as dark with street lights not functioning.

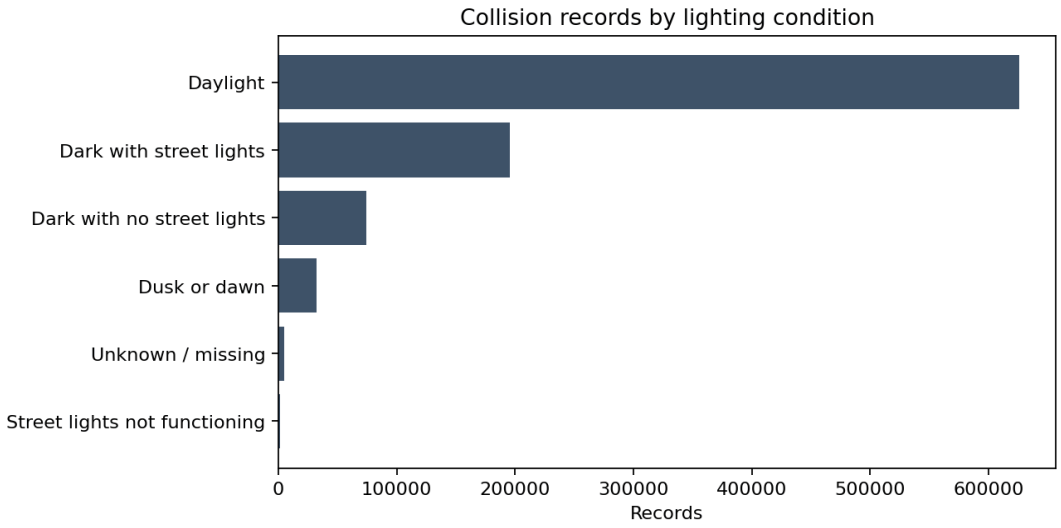


Figure 3. Collision records by lighting condition.

6.5 Road-Surface Conditions

Dry surfaces account for 845,612 records and wet surfaces for 76,831. Snowy and slippery surfaces have smaller counts (4,122 and 1,048), while 8,141 records have no standardized road-surface value.

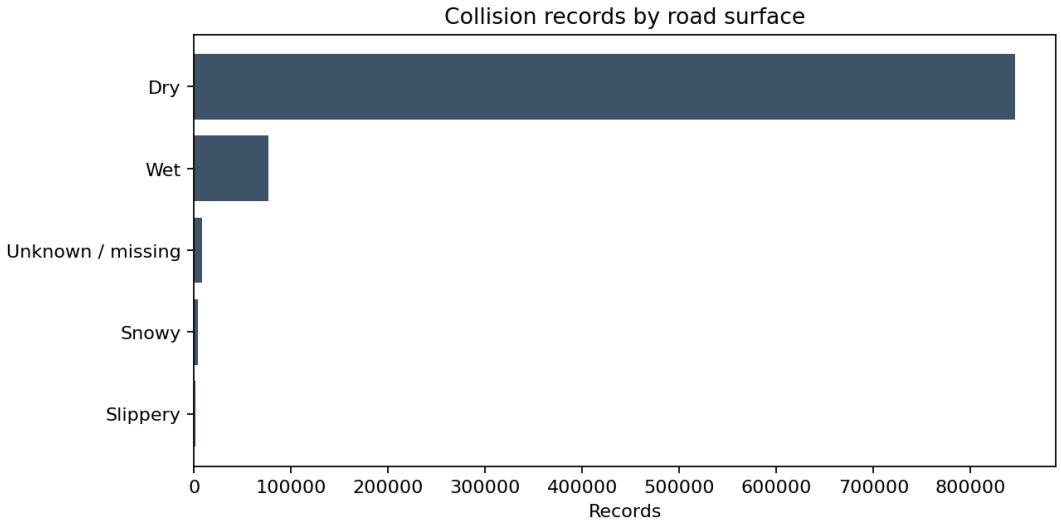


Figure 4. Collision records by road surface.

6.6 Weekday Patterns

Friday has the highest weekday count (154,539), followed by Thursday (138,710) and Wednesday (137,287). Sunday has the lowest count (109,078). Weekday totals do not estimate collision probability per trip.

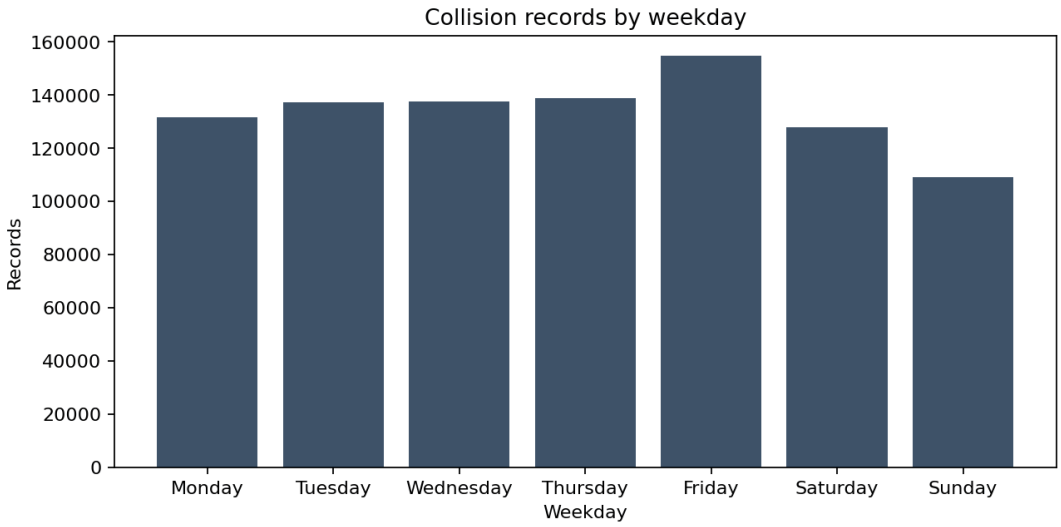


Figure 5. Collision records by weekday.

6.7 County Distribution and Geography

Los Angeles County has the highest count (284,088), followed by Orange (72,038), San Bernardino (56,737), San Diego (53,103) and Riverside (48,686). The geographic scatter plot includes 40,247 valid coordinate points after sampling. With approximately 71.5% missing latitude and longitude values, the map represents only a subset of collision records. County counts are absolute recorded volumes, not per-capita or per-vehicle-mile risk. Cross-county comparisons require suitable exposure denominators.

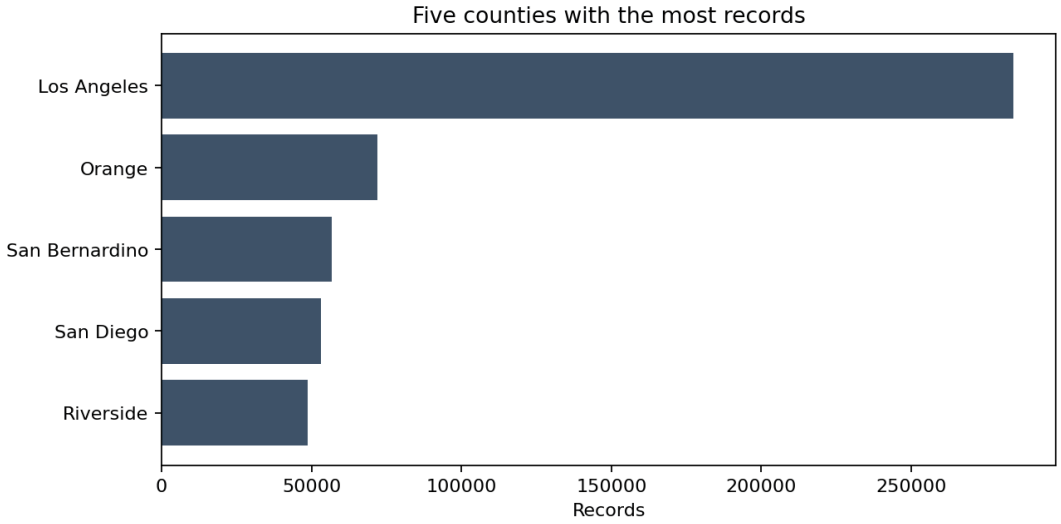


Figure 6. Five counties with the most records.

6.8 Monthly and Hourly Trends

October is the highest-count month (83,272), followed by March (82,164). The lowest month in the displayed summary is June (74,760). At the hourly level, 17:00 is the peak (73,252), followed by 15:00 (70,855) and 16:00 (67,431).

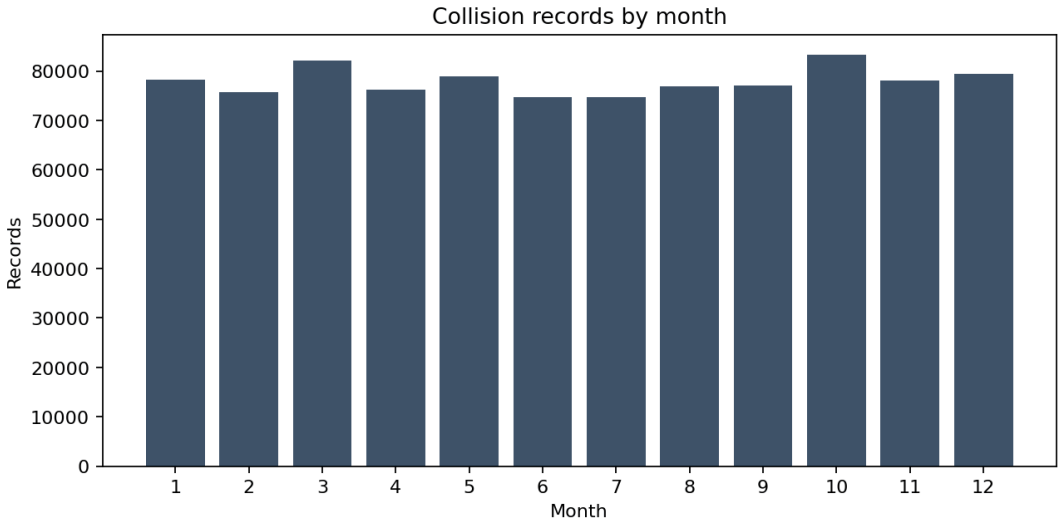


Figure 7. Collision records by month.

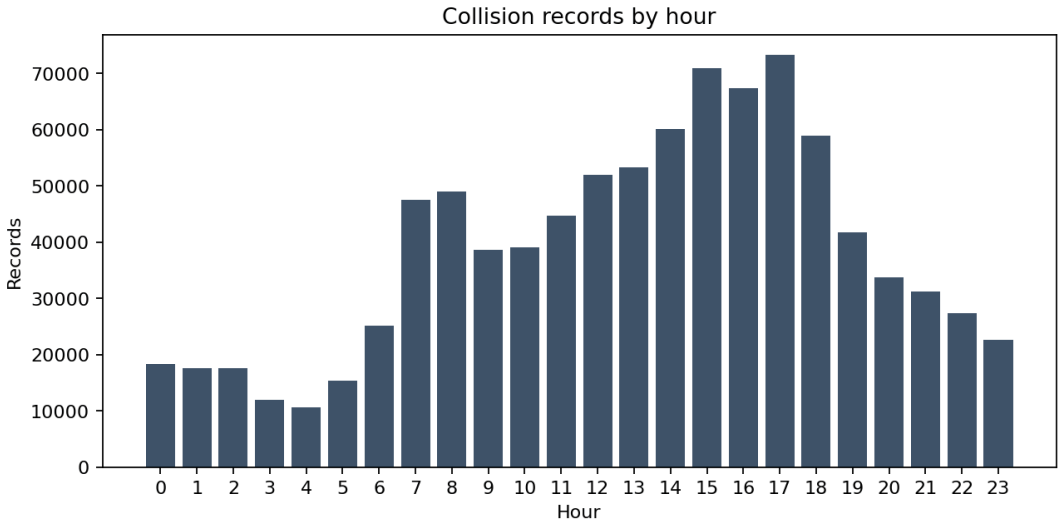


Figure 8. Collision records by hour.

6.9 Yearly Trend

Yearly counts are highest in 2002 (53,708) among the displayed years and generally decline through the early-to-mid 2010s before rising again in 2015–2018. The 2020 total is 36,020 and the 2021 total is 13,198; 2021 is partial, so the final value is not directly comparable with complete calendar years.

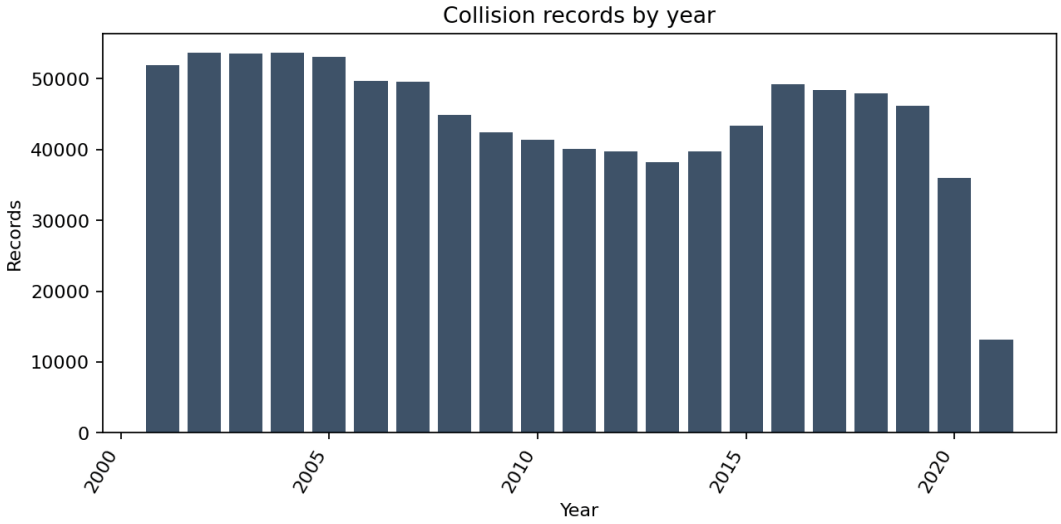


Figure 9. Collision records by year.

7. ETL Query Results

Query	Result from notebook
Top five counties	Los Angeles (284,088); Orange (72,038); San Bernardino (56,737); San Diego (53,103); Riverside (48,686)
Month with highest count	October — 83,272 records
Most common weather	Clear — 769,905 records
Peak hour by count	17:00 — 73,252 records
Most common road surface	Dry — 845,612 records
Most common lighting condition	Daylight — 625,553 records
Fatal-collision percentage	Not estimable from the collision-level killed_victims field, which is zero for all non-null records; validated victim-level linkage is required.

8. Key Findings and Practical Implications

1. Prioritize high-volume locations for deeper assessment, then normalize by traffic exposure before allocating resources.
2. Use Friday and afternoon/evening peaks as starting points for examining congestion, signal timing, speed patterns and enforcement coverage.
3. Assess lighting and road conditions with rates and severity shares, not raw counts alone.
4. Improve geospatial completeness to support corridor-level hotspot analysis.
5. Validate fatality measures through a confirmed linkage between collision and victim records.
6. Monitor yearly patterns while accounting for partial-year coverage and reporting changes.

9. Limitations

The analysis is descriptive and based on the supplied extract. It does not measure risk per trip or distance travelled and does not establish causation. Missing location data limits map completeness, the final year is partial, and a reliable fatal-collision percentage requires validated linkage to victim-level fatality records. Results should be interpreted in the context of source coding and reporting practices.

10. Conclusion

The project demonstrates a Spark-based ETL and exploratory-analysis workflow for a large California traffic-collision extract. It summarizes nearly 936,000 cleaned collision-level records and identifies patterns across severity, county, weekday, month, hour, weather, road surface and lighting. Property-damage-only records dominate the severity distribution; Los Angeles County has the highest recorded count; October is the peak month; Friday is the busiest weekday; and 17:00 is the peak hour by raw count. These findings provide a descriptive baseline for traffic-safety planning. Stronger risk assessment would combine improved location completeness, validated fatality linkage and exposure-adjusted measures.

Appendix: Summary Metrics

Metric	Value
Cleaned collision-level records	935,764
Property damage only	570,460 (61.0%)
Pain	226,486 (24.2%)
Other injury	116,680 (12.5%)
Severe injury	22,138 (2.4%)
Friday records	154,539
October records	83,272
17:00 records	73,252
Valid coordinate points plotted	40,247